$d\Omega = \underbrace{-SdT - PdV}_{dF} - \sum_i N_i d\mu_i \quad ; \quad \Omega(T,V,\mu_i)$



Const V_{tot} ; $\Omega^{tot} = \Omega^\alpha + \Omega^\beta + \Omega$

Free energy per volume in bulk α $\omega^\alpha V^\alpha$

$d\Omega^t = d\Omega^\alpha + d\Omega^\beta + d\Omega$

$d\Omega^\alpha = -S^\alpha dT - P dV^\alpha - \sum_i N_i^\alpha d\mu_i \quad ; \quad d\Omega^\beta = -S^\beta dT - P dV^\beta - \sum_i N_i^\beta d\mu_i$

$T^\alpha = T^\beta = T^{interface}$
$P^\alpha = P^\beta = P^{int}$
$\mu_i^\alpha = \mu_i^\beta = \mu_i^{int}$

How N_i at V_i?

$N_{in} = N_t - \rho_m V_{m,bulk} - N_m^{vapor}$

excess amount of interfacial material

Including interfacial term : $d\Omega^t = -S^t dT - P\,dV^t + \gamma dA - \sum_i N_i^t d\mu_i$

work associated w/ changes in interfacial area

Also: $S^t = S^\alpha + S^\beta + S$; $N_i^t = N_i^\alpha + N_i^\beta + N_i$; $V^t = V^\alpha + V^\beta$ (Couple of atomic layers thick)

$S^\alpha V^\alpha$ (per volume)

Boundary

Often placed where there is a maximum conc. gradient

$d\Omega = \left(-S^t dT - P dV^t + \gamma dA - \sum N_i^t d\mu_i\right) + \left(S^\alpha dT + P dV^\alpha + \sum N_i^\alpha d\mu_i\right) + \left(S^\beta dT + P dV^\beta + \sum N_i^\beta d\mu_i\right)$

$d\Omega = -\underbrace{(S^t - S^\alpha - S^\beta)}_{S} dT - P d(\underbrace{V^t - V^\alpha - V^\beta}_{0}) + \gamma dA - \sum_i (N_i^t - N_i^\alpha - N_i^\beta) d\mu_i$

No excess interfacial volume

$\therefore \boxed{d\Omega = -SdT + \gamma dA - \sum_i N_i d\mu_i}$ All quantities are excess interfacial quantities $\Omega = \Omega(T, A, \mu_i)$

Euler Integration → $-\Omega = \gamma A$ → $\boxed{\gamma = \frac{-\Omega}{A}}$

Surface tension is excess Ω per unit area

; Aside: $-\Omega = \gamma A$

$d\Omega = \gamma dA + A d\gamma$

$A d\gamma = -SdT - \sum_i N_i d\mu_i$

$d\gamma = -\frac{S}{A} dT - \sum_i \frac{N_i}{A} d\mu_i$

Gibbs-Duhem for interfaces → Relates to Gibbs-Duhem Isotherm $\gamma = \gamma(T, \mu_i)$

Calculate γ at given $\{T, \mu_i\}$ and find structure w/ lowest γ

↳ Surface phase diagram

Idea: Calculate $\frac{-\Omega}{A}(X)$, using DFT + stat mech, assuming surface is in equilibrium w/ bulk phases: (b/c $\mu_i = \mu_i^{gas}$, $\mu_i = \mu_i^{bulk\ metal}$)

$\boxed{\gamma = (U - TS - \Sigma_i \mu_i N_i)\frac{1}{A}}$; Find lowest γ for desired T, μ_i

XS interfacial internal energy ; XS entropy ; XS moles of i

Algorithm: ① U can be approximated in many ways: (potential energy for atoms)

+ Ignore kinetic of surface molecules, and assume $U = E_{DFT}$

$\gamma = \frac{E_{DFT} - \rho_{bulk} V^\alpha}{A}$ (Bulk energy density)

} Often people ignore this XS aspect of surface energy

↳ phase diagrams

Better + U ≈ E_{DFT} + ZPE

↳ zero point energy (Lowest vibrational energy state for molecules)

Note: typically neglect ZPE of surface itself

Term project ideas:
+ Methods of app for Q
+ phonon interest in ZPE

Best + Calculate $Q = \sum_j e^{-\beta E_j}$ → states of surface (have to make approximations since infinite) and use $U = \frac{-\partial \ln Q}{\partial \beta}$

↑ partition function

② TS excess surface entropy approximations:

+ Assume TS = 0 (Surfaces are not bulk phases → reduced dimensionality Change decrease in entropy)

First approximation

+ Adsorbates: count all possible arrangements of adsorbate "i", then $S_i = k_B \ln \Omega$ (No. of states)

↳ Configurational entropy : mean field treatment

+ Partition function : $S = k_B \ln Q - \frac{\partial \ln Q}{\partial \beta}$ (Calculated for Stat mech)

↳ Adsorbates: count all possible arrangements of adsorbate i, then $\nu_i = k_B \ln \Omega$

↳ Configurational entropy: mean field treatment

+ Partition function: $S = k_B \ln Q - \frac{1}{T}\frac{\partial \ln Q}{\partial \beta}$ (Calculated for Stat. mech)

+ Use empirical correlations for S_i of surface adsorbates.

③ $\{\mu_i\}$

+ Leave these as undetermined independent parameters



+ Evaluate μ_i by equilibrium w/ a bulk phase $\rightarrow \mu_i = \mu_i^{gas}$

↳ Use measured data: $\mu_{gas}(T,P) = \mu_{gas}(T,P^\circ) + k_B T \ln\left(\frac{P}{P^\circ}\right)$ (i.g.)

$= h_{gas}(T,P^\circ) - TS_{gas}(T,P^\circ) + k_B T \ln\left(\frac{P}{P^\circ}\right)$

$= \underbrace{\left[h_{gas}(T,P^\circ) - h_{gas}(0,P^\circ)\right]}_{\int_0^T c_p\,dT \text{ @ } P^\circ} + \underbrace{\left[h_{gas}(0,P^\circ) - E^{DFT}_{gas}\right]}_{\text{Only KE at 0K is the zero point energy}} + \underbrace{\left[E^{DFT}_{gas} - TS_{gas}(T,P^\circ)\right]}_{\text{Potential energy from DFT calc. of gas phase}} + k_B T \ln\left(\frac{P}{P^\circ}\right)$

(absolute vals. from tables)

} Adding & subtracting various reference states

$h_{gas}(0,P^\circ) - E^{DFT}_{gas} = ZPE$

$\int_0^T c_p\,dT$ @ P° → entropolate, assume constant, etc.

↳ $\mu_{gas} = h_{gas}(T,P) - TS_{gas}(T,P)$

↓

$h_{gas}(T,P) = U_{gas}(T,P) + \underbrace{k_B T}_{\text{ideal gas}}$; Find U_{gas} w/ : (i) Equipartition theorem (high T)

Total DOF is $3N$ (the same)

+ Each translational mode (3 total) gives $\frac{1}{2}k_B T$
+ Each vibrational mode ($3N-6$ non-linear, $3N-5$ linear) gives $k_B T$
+ Each rotational mode (3 non-linear, 2 linear) gives $\frac{1}{2}k_B T$

$\Rightarrow U_{gas} = \frac{3}{2}k_B T + (3N-6)k_B T + E^{DFT}_{gas}(\text{non-linear})$

$= \frac{5}{2}k_B T + (3N-5)k_B T + E_{DFT,gas}(\text{linear})$

(ii) Real statistical mechanics for U_{gas} (& S_{gas}):

Vibrational: $Q_{vib} = \prod_i \frac{e^{-\frac{h\nu_i}{2}\beta}}{1 - e^{-h\nu_i \beta}}$ (per molec.) ; ν_i = vibrational freq. (measured or calc. from) ; i = all normal modes of molecule

Rotational: $Q_{rot} = \frac{\sqrt{\pi}}{\sigma}\sum_{j=1}^{3}\left(\frac{2I_j}{\hbar^2\beta}\right)^{1/2}$ non-linear ; σ → symmetry factor, $\beta = \frac{1}{k_B T}$, $I_{1,2,3}$ → principle moments of inertia

$Q_{rot} = \frac{I}{\sigma\beta\hbar^2}$ (linear)

Q

Translational: $Q_{trans} = \left(\frac{m}{2\pi\beta\hbar^2}\right)^{3/2} V$ (for 1 molecule) ; V = total volume of system

$U = \langle E \rangle = \frac{\sum E_i e^{-\beta E_i}}{Q}$ ($e^{-\beta E_i}$: probability of system in state)

$= -\frac{\partial \ln Q}{\partial \beta}$

$= -\frac{1}{Q}\frac{\partial Q}{\partial \beta}$

$U = -\frac{\partial \ln Q}{\partial \beta}$